

DEB KANTI BARUA

Senior Backend Engineer | Team Lead | C# / .NET | Java | Microservices

Dhaka, Bangladesh | 01836309702 | deb.kanti@icloud.com

SUMMARY

Senior software engineering professional and Solution Architect with 13+ years of experience in backend engineering, microservices, API design, security, enterprise integration, and technical leadership. Hands-on experience with C#, ASP.NET Core, .NET, Java, Spring Boot, REST, gRPC, and EF Core across greenfield systems and modernization initiatives. Experienced in building high-performance inter-service communication, secure APIs, regulatory/e-invoicing compliance, payment integration, and enterprise platforms. Published research in deep-learning-based human trajectory imputation for pedestrian safety.

CORE COMPETENCIES

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| • Backend Engineering | • C# / .NET |
| • Java / Spring Boot | • Microservices Architecture |
| • System Design | • RESTful API Design |
| • gRPC & Inter-Service Communication | • Performance Optimization |
| • Security & Cryptography | • Enterprise Integration |
| • Regulatory / E-Invoicing Compliance | • Payment Integration |
| • Design Patterns & SOLID Principles | • Technical Leadership |
| • Code Review & Engineering Standards | • Legacy System Modernization |
| • Requirements Engineering | • BRD & SRS Development |
| • Agile & Cross-functional Collaboration | • R&D; |

TECHNICAL LEADERSHIP

- **Team Leadership:** Led engineering teams of 5–8 developers across enterprise, government, OTA, POS/ERP, and simulation projects while remaining hands-on with backend engineering and architecture.
- **Hands-on Leadership:** Combined technical leadership with backend development, architecture, code reviews, system design, performance optimization, and production problem solving.
- **Mentorship & Engineering Standards:** Guided developers on architecture, design patterns, coding practices, inter-service communication, and implementation standards.

PROFESSIONAL EXPERIENCE

Solution Architect — Akij iBos Feb 2025– Present

- Designed and implemented microservices architecture for an OTA platform.
- Defined and maintained architectural standards and reusable engineering patterns.
- Implemented repository/service-layer patterns with extensible generic CRUD support.
- Developed a reusable framework that enables developers to easily implement gRPC-based inter-service communication across services.
- Led architectural decisions, R&D, and technical problem solving; performed comprehensive code reviews and provided technical guidance to a team of 8 engineers.
- Delivered backend services for B2B partner approval, notifications, configuration, commission/markup, booking, and ticketing.
- Transferred substantial business logic from a WinUI 3 desktop POS application to ASP.NET Core services, significantly reducing the desktop application's workload and footprint.
- Optimized POS invoice-generation performance using SemaphoreSlim.
- Designed and implemented a Saudi Arabia e-invoicing compliance layer involving cryptography and security.
- Prepared BRD and SRS for a ZATCA-equivalent e-invoicing/compliance platform for another country.
- Integrated ESR payment services with POS, streamlining the payment flow.
- Added sales and demand forecasting to an ERP using the open-source Prophet forecasting model.
- Worked across backend architecture, modernization, performance, security, compliance, payment integration, and analytics.

Technology Lead — Cokcreates Ltd Mar 2019 – Aug 2020

- Core microservices architecture and engineering team for a Spring Boot Government Resource Planning (GRP) platform.
- Worked on a government platform supporting approximately 200 concurrent users.
- Led the Common Services team of 8 engineers.
- Delivered office organogram, posting/transfer, committee formation, and notification infrastructure covering SMS, email, and push notifications.
- Provided technical guidance for PIMS, Attendance & Leave, Disciplinary, and Training systems.
- Drove consistent architecture and implementation practices across the platform.

Head of IT — E-Horizon IT Nov 2018 – Feb 2019

- Solely developed a Media Archive System for a government counter-terrorism agency.
- Implemented KPI analytics and news-archiving workflows.
- Worked with Express.js, raw PHP, jQuery, and Bootstrap.
- Provided technical and delivery leadership for broader IT initiatives.

Senior Software Engineer — R&D; — Akij Group Apr 2017 – Oct 2018

- Modernized a legacy HRMS/Payroll system into JWT-secured REST APIs.
- Implemented OpenID authentication and Serilog structured logging.
- Performed architectural refactoring of Accounts and Inventory modules.
- Conducted R&D; and integrations using Syncfusion PDF, Apache POI, Siemens PLC Web Server, and RFID SDKs.
- Delivered multi-stack solutions across enterprise applications and integrations.

Senior Project Manager — REVE Systems Jan 2015 – Apr 2017

- Led development teams of 5–6 engineers while remaining hands-on technically.
- Delivered a KPI Email Support System for Grameenphone.
- Worked on government messaging and file-sharing platform a2i Infosarkar.
- Worked on VAWCPS and a Software-defined Radio solution in C++ for a government intelligence agency.
- Worked with Java, Struts, JAX-RS, jQuery, Bootstrap, Metronic, and enterprise integrations.

Team Lead — Hi-Tech Bangla Bangladesh Ltd Dec 2012 – Apr 2014

- Led a team of 6–8 engineers developing a Radar Control Simulator for Bangladesh Air Force FCTU.
- Worked on aircraft interception and war-gaming simulation.
- Implemented real-time visualization using Java Swing and OpenStreetMap.
- Provided technical leadership across simulation software development.

Senior Software Engineer — Advanced Software Development Dec 2005 – Jan 2009

- Developed enterprise software for Standard Chartered, HSBC, RSF, and BAT.
- Worked with Java, Spring, JSP, Servlets, and ASP.NET.

TECHNICAL SKILLS

Languages: Java, C#, C++, PHP, JavaScript

Backend & Frameworks: Spring Boot, Spring Security, Spring Cloud, ASP.NET Core, .NET, Entity Framework Core, JAX-RS, Servlets, JSP, Struts, Express.js

Architecture & Engineering: Microservices, REST APIs, gRPC, System Design, Repository Pattern, Unit of Work, Generic CRUD Architecture, Design Patterns, SOLID Principles

Security & Compliance: JWT, OpenID, Authentication & Authorization, Cryptography, Secure API Design, E-Invoicing Compliance

Data & Forecasting: SQL Server, Forecasting & Demand Planning, Prophet

Tools & Other Technologies: Git/GitLab, Serilog, Apache POI, Syncfusion PDF, RFID SDK, Siemens PLC Web Server, OpenStreetMap, Java Swing

PUBLICATIONS

PTIN: Enriching Pedestrian Safety with an LSTM-GRU-Transformer Based Trajectory Imputation System for Autonomous Vehicles

Conference Paper — 26th International Conference on Computer and Information Technology (ICCIT), IEEE, 2023. Research on deep-learning-based human trajectory imputation for pedestrian safety in autonomous vehicles.

Human trajectory imputation model: A hybrid deep learning approach for pedestrian trajectory imputation

Journal Article — Applied Sciences, Vol. 15, Issue 2, MDPI, 2025. Research proposing a hybrid deep-learning approach for human/pedestrian trajectory imputation, evaluated using the inD dataset.

EDUCATION

B.Sc. in Computer Science & Engineering — Ongoing — United International University Oct 2020 Present

131 credits completed; 6 credits remaining; CGPA: 3.4

B.Sc. in Computer Science & Engineering — Completed up to 3rd Semester — Bangladesh University of Engineering and Technology (BUET)